Finite Exponential Moment Does Not Force Finite-Time Ornstein–Uhlenbeck Poincaré Regularization

Candidate full counterexample packet

arXiv:2504.08658v2, Section 3.2.2

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Status. Candidate full negative answer, subject to expert review. We construct a normalized centered nonnegative $v \in H^1(\mathbb{R}, d\gamma)$ with a finite exponential moment such that $|w(t)|^2 d\gamma$ fails every Poincaré inequality at every finite Ornstein–Uhlenbeck time t.

1 Source question

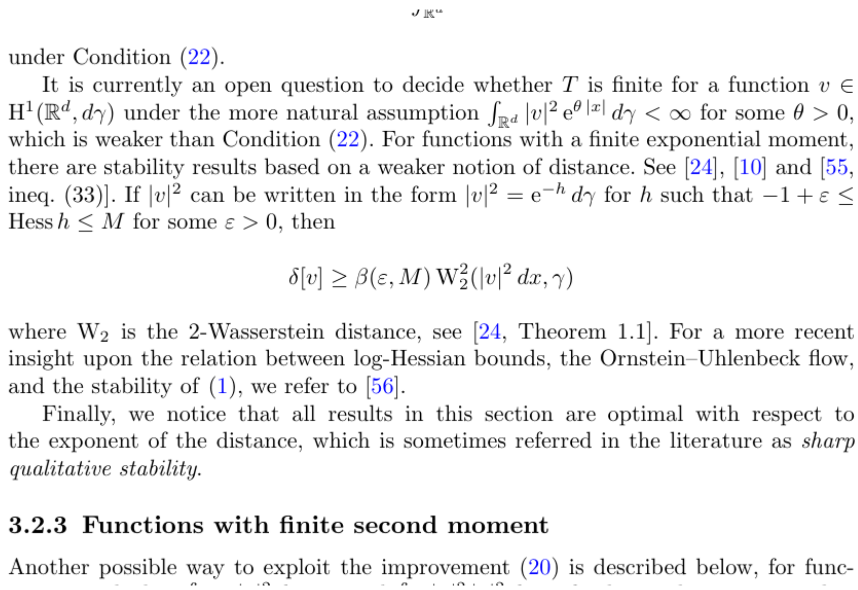
Let $d\gamma(x) = (2\pi)^{-1/2} e^{-x^2/2}\, dx$. In Section 3.2.2, Brigati–Dolbeault–Simonov [1, p. 17] consider the nonlinear form of the Ornstein–Uhlenbeck evolution

$$\partial_t w = w'' + \frac{|w'|^2}{w} - xw', \qquad w(0) = v. \tag{1}$$

They ask whether the assumption

$$v \in H^1(\mathbb{R}, d\gamma), \qquad \int_{\mathbb{R}} |v|^2 e^{\theta|x|}\, d\gamma < \infty \quad \text{for some } \theta > 0 \tag{2}$$

forces the existence of a finite time T at which $|w(T)|^2 d\gamma$ satisfies a Poincaré inequality. Their surrounding discussion takes the density to be normalized and centered; our example has both properties.



under Condition (22).

It is currently an open question to decide whether T is finite for a function $v \in \mathrm{H}^1(\mathbb{R}^d, d\gamma)$ under the more natural assumption $\int_{\mathbb{R}^d} |v|^2\, e^{\theta\,|x|}\, d\gamma < \infty$ for some $\theta > 0$, which is weaker than Condition (22). For functions with a finite exponential moment, there are stability results based on a weaker notion of distance. See [24], [10] and [55, ineq. (33)]. If $|v|^2$ can be written in the form $|v|^2 = e^{-h}\, d\gamma$ for h such that $-1 + \varepsilon \le \operatorname{Hess} h \le M$ for some $\varepsilon > 0$, then

$$\delta[v] \ge \beta(\varepsilon, M)\, \mathrm{W}_2^2(|v|^2\, dx, \gamma)$$

where W_2 is the 2-Wasserstein distance, see [24, Theorem 1.1]. For a more recent insight upon the relation between log-Hessian bounds, the Ornstein–Uhlenbeck flow, and the stability of (1), we refer to [56].

Finally, we notice that all results in this section are optimal with respect to the exponent of the distance, which is sometimes referred in the literature as *sharp qualitative stability*.

3.2.3 Functions with finite second moment

Another possible way to exploit the improvement (20) is described below, for func-

Figure 1: The open question in the current source PDF, page 17.

2 Construction and statement

Put

$$a_k = 2^k, \qquad Z = \sum_{k=1}^{\infty} e^{-a_k}, \qquad p_k = Z^{-1} e^{-a_k}.$$

Let $\nu_m = N(m,1)$ and define the symmetric probability measure

$$\mu = \sum_{k=1}^{\infty} \frac{p_k}{2}\big(\nu_{a_k} + \nu_{-a_k}\big). \tag{3}$$

Relative to $\gamma = N(0,1)$, the translate ν_m has density

$$g_m(x) = \exp(mx - m^2/2).$$

Thus $\mu = f\gamma$, where

$$f(x) = \sum_{k=1}^{\infty} p_k e^{-a_k^2/2}\cosh(a_k x), \qquad v = \sqrt{f}. \tag{4}$$

Theorem 1. *The function v in (4) is nonnegative, normalized, centered, and belongs to $H^1(\mathbb{R}, d\gamma)$. It satisfies (2) for every $0 < \theta < 1$. Nevertheless, for every finite $t \ge 0$, the measure $|w(t)|^2 d\gamma$ generated by (1) has no positive Poincaré constant. In particular the finite-time question of [1] has a negative answer.*

3 Regularity and exponential moment

Normalization and centering follow at once from the probabilistic mixture (3). If X has law μ, then conditionally on the index and sign it has the form $X = \pm a_k + G$, where $G \sim N(0,1)$. Hence for $0 < \theta < 1$,

$$\begin{aligned}\mathbb{E}e^{\theta|X|} &\le \mathbb{E}e^{\theta|G|}\sum_{k=1}^{\infty} p_k e^{\theta a_k} \\ &= \frac{\mathbb{E}e^{\theta|G|}}{Z}\sum_{k=1}^{\infty} e^{-(1-\theta)2^k} < \infty.\end{aligned}$$

It remains to verify the nontrivial H^1 requirement. Index all signed components by i, writing $f = \sum_i q_i g_{m_i}$. Since $g'_{m_i} = m_i g_{m_i}$, Cauchy–Schwarz gives pointwise

$$\frac{|f'|^2}{f} = \frac{|\sum_i q_i g'_{m_i}|^2}{\sum_i q_i g_{m_i}} \le \sum_i q_i \frac{|g'_{m_i}|^2}{g_{m_i}} = \sum_i q_i m_i^2 g_{m_i}. \tag{5}$$

The series and its derivatives converge locally uniformly; alternatively, (5) follows first for finite mixtures and then by monotone approximation. Integrating and using $\int g_m\, d\gamma = 1$ yields

$$4\int_{\mathbb{R}} |v'|^2\, d\gamma = \int_{\mathbb{R}} \frac{|f'|^2}{f}\, d\gamma \le \sum_{k=1}^{\infty} p_k a_k^2 = \frac{1}{Z}\sum_{k=1}^{\infty} 4^k e^{-2^k} < \infty.$$

Together with $\int v^2 d\gamma = 1$, this proves $v \in H^1(\gamma)$.

4 Exact evolution of the mixture

Let P_t be the Ornstein–Uhlenbeck semigroup and set $r = e^{-t}$. Mehler's formula gives the exact identity

$$P_t g_m(x) = \exp(rmx - r^2m^2/2) = g_{rm}(x). \tag{6}$$

Consequently $\mu_t = (P_t f)\gamma$ is the symmetric mixture

$$\mu_t = \sum_{k=1}^{\infty} \frac{p_k}{2}\big(N(ra_k, 1) + N(-ra_k, 1)\big). \tag{7}$$

If $w(t) = \sqrt{P_t f}$, then $\partial_t(w^2) = (\partial_x^2 - x\partial_x)(w^2)$, and division by $2w$ gives (1). Thus (7) is exactly the measure appearing in the source question.

5 Vanishing Poincaré Rayleigh quotients

Fix an arbitrary finite t, so $r > 0$, and write $b_j = ra_j$. For $k \geq 2$ set

$$L_k = \frac{b_{k-1} + b_k}{2}, \qquad U_k = \frac{b_k + b_{k+1}}{2}.$$

Choose a smooth cutoff h_k with values in $[0, 1]$ which is one on $[L_k + 1, U_k - 1]$, zero outside $[L_k - 1, U_k + 1]$, and satisfies $|h_k'| \leq 1$. Such cutoffs exist for all sufficiently large k.

Under the component $N(b_k, 1)$, the plateau endpoints are at signed distances

$$-\frac{r(a_k - a_{k-1})}{2} + 1 = -r2^{k-2} + 1, \qquad \frac{r(a_{k+1} - a_k)}{2} - 1 = r2^{k-1} - 1$$

from its mean. Therefore this component assigns the plateau probability at least $1/2$ for all large k. Hence

$$\mu_t\{h_k = 1\} \geq \frac{p_k}{4}.$$

On the other hand, at least half of every negative-mean Gaussian lies to the left of $L_k - 1$ for large k. The total negative-component weight is $1/2$, so $\mu_t\{h_k = 0\} \geq 1/4$. Using $\operatorname{Var}_{\mu_t} h = \frac{1}{2} \iint |h(x) - h(y)|^2 d\mu_t(x) d\mu_t(y)$, we obtain

$$\operatorname{Var}_{\mu_t}(h_k) \geq \frac{p_k}{16}. \tag{8}$$

The derivative is supported in the two intervals of radius one centered at L_k and U_k. Every component mean $\pm b_j$ is at distance at least $r2^{k-2}$ from their union's nearer midpoint. A standard normal assigns an interval of length two at midpoint-distance $d > 1$ mass at most $2(2\pi)^{-1/2} e^{-(d-1)^2/2}$. Summing all mixture weights therefore gives

$$\int |h_k'|^2 d\mu_t \leq \frac{4}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(r2^{k-2} - 1\right)^2\right]. \tag{9}$$

Combining (8), (9), and $p_k = Z^{-1} e^{-2^k}$,

$$\frac{\int |h_k'|^2 d\mu_t}{\operatorname{Var}_{\mu_t}(h_k)} \leq \frac{64Z}{\sqrt{2\pi}} \exp\left[2^k - \frac{1}{2}\left(r2^{k-2} - 1\right)^2\right] \longrightarrow 0. \tag{10}$$

The cutoff can be chosen directly smooth as above, or obtained by smoothing a piecewise linear cutoff without changing the estimates. Thus no constant $C_P > 0$ can make

$$\int |\phi'|^2 d\mu_t \geq C_P \operatorname{Var}_{\mu_t}(\phi) \quad \text{for all } \phi \in C_c^\infty(\mathbb{R}).$$

Because the finite time t was arbitrary, this proves Theorem 1.

6 Scope and verification

The counterexample is one-dimensional, which suffices to refute the dimension-free question. It exploits infinitely many increasingly remote bottlenecks. Its finite exponential moment is strictly weaker than the source's Gaussian two-point moment condition (22), which this mixture does not satisfy, so it does not conflict with the cited positive theorem.

The formula in (10) was checked numerically on a logarithmic scale for several finite times. A bounded search through 11 August 2026 of the run registry, the current arXiv source and published DOI, exact phrases, and Gaussian-mixture/Poincaré keywords found no later resolution or this construction. This is not a claim of priority.

Human-review recommendation. Verify the transition-strip bound in (9). Also check the identity $|w(t)|^2 d\gamma = \mu_t$. The obstruction is then immediate.

References

[1] G. Brigati, J. Dolbeault, and N. Simonov, *Logarithmic Sobolev inequalities: a review on stability and instability results; logarithmic uncertainty principle*, La Matematica 5 (2026), article 5; arXiv:2504.08658v2.